

The Solar System

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS The **Sun** is our closest star. **Eight planets** orbit it in order from the Sun: **Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune**. Earth is the **third** planet from the Sun. Stars look different in brightness mostly because of their **distance** from us.

Round 1 — Retrieve It

1. What is the closest star to Earth?

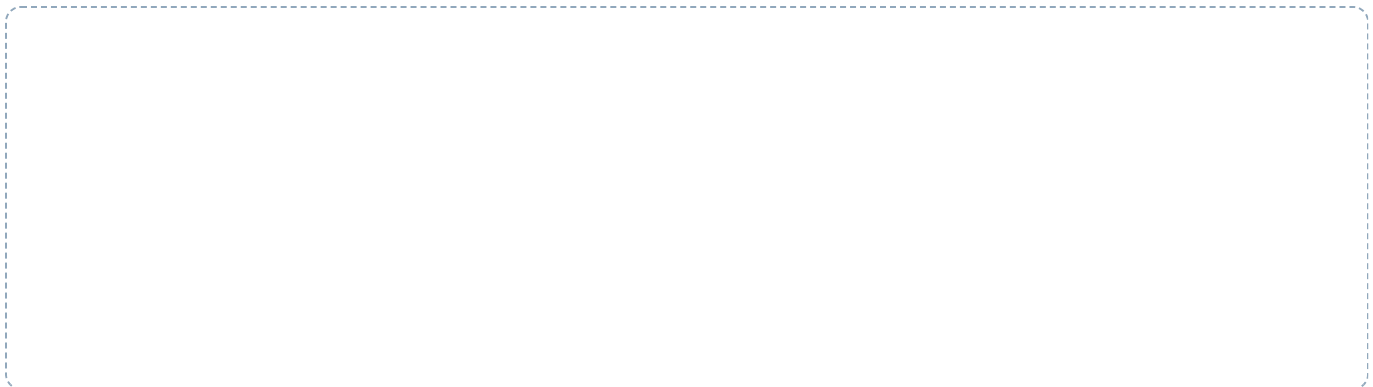
2. How many planets orbit the Sun?

3. Which planet is third from the Sun?

4. Why do some stars look brighter than others?

Round 2 — Sketch It

Draw the Sun and the eight planets in order. Circle Earth and label it "third from the Sun."



The Solar System — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. Name the first four planets in order starting from the Sun.

2. The Sun looks much bigger and brighter than the stars we see at night. Why is that, even though the Sun is a star too?

3. Which planet comes right after Earth as you move away from the Sun, and which comes right before it?

Round 4 — Reason from Evidence

Two stars are really the same brightness, but one looks much dimmer from Earth. What is the most likely reason, and how do you know?

The Solar System — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain what a solar system is. Tell what is at the center and what orbits around it.

C**Connect**

Connect distance to brightness. Why does the Sun look so bright to us while other stars look like tiny dots?

E**Extend**

Make up a sentence or trick to help you remember the eight planets in order. Then list the planets it helps you recall.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — The Solar System

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. The Sun.
2. Eight planets.
3. Earth.
4. Mostly because of distance; stars that are farther away look dimmer, and closer ones look brighter.

Sketch It — look for

Sun with eight planets in order (Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune), Earth circled and labeled as third from the Sun.

Apply It

1. Mercury, Venus, Earth, Mars.
2. The Sun looks bigger and brighter because it is much closer to us than any other star.
3. After Earth comes Mars; before Earth comes Venus.

Reason from Evidence — look for

The dimmer star is most likely much farther away. Brightness we see depends a lot on distance, so a more distant star looks fainter even if it gives off the same amount of light.

ACE — Extend look-for

A memory trick (such as a sentence where each word starts with M, V, E, M, J, S, U, N) that correctly lists the eight planets in order.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.